DSA REFERENCE

Monotonic Stack Failure Modes

Common wrong turns, debugging cues, and complexity traps.

Keep unresolved candidates in monotonic order; a new value resolves every dominated candidate once.

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DSA / MONOTONIC-STACK

DSA

Monotonic Stack Failure Modes

CORE IDEA

Keep unresolved candidates in monotonic order; a new value resolves every dominated candidate once.

Recognition cues

+ next greater or smaller
+ span
+ histogram area
+ nearest boundary

Implementation sequence

+ store indices
+ choose strict or non-strict comparison
+ resolve during pops
+ flush with a sentinel when necessary

Invariant

Stack indices remain monotonic by value and each stored index is still waiting for its resolving boundary.

Worked reasoning

For next greater element, pop smaller values when a larger value arrives and assign the current index as their answer.

Complexity

O(n) time because each index is pushed and popped at most once; O(n) space.

Failure modes

- [ ] wrong equality rule
- [ ] storing values without positions
- [ ] missing final flush
- [ ] confusing previous and next boundaries

Practice ladder

Monotonic Stack Failure Modes: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Daily Temperatures**	State the cue, invariant, edge case, and proof before implementation.
2	**Next Greater Element I**	State the cue, invariant, edge case, and proof before implementation.
3	**Largest Rectangle in Histogram**	State the cue, invariant, edge case, and proof before implementation.
4	**Car Fleet**	State the cue, invariant, edge case, and proof before implementation.
5	**Sum of Subarray Minimums**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

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